

The Utility Problem

As electricity distribution grids are becoming more modern, new operational measurement data streams are being created in the grid on a daily basis through the integration of various time series generation devices. This time series data presents a huge potential for insights and situational awareness. To get these potent data out of the swell of time series, one of the adapted techniques is setting the thresholds for which only desired data are highlighted. Such thresholds then generate alerts and warnings which can be analyzed further.

Setting thresholds for displaying the selected variables is very important, as it might indicate possible problems arising in the grid. When alerts and warning data are analyzed, an important part of the process is often forgotten: proper data visualization. Results in tables are often not as clear as results on a map - for example, geospatial coordinates tell us nothing, or it is very difficult to know all the streets in all the cities where a utility operates. Map results also offer a new perspective on the problem, and often the problem can be solved more easily since it provides the context of the space and relation to other geospatial elements. On the map, areas of the cluster of values where a threshold has been exceeded are visible at a glance and it is easier to identify it for further detailed investigation.

Another advantage of geospatial alerts and warnings data becomes apparent when there are many results as a map chart is much less overcrowded than other types of charts. While a bar chart with 100 or more results would be overcrowded, displaying the same data on a map is very clear, digestible and pleasing to the eye. Of course, map visualizations cannot be used for all results, but they are more suitable for this type of display. Another important parameter is the level of detail we are able to perform with the analysis. Analyzing the whole grid without breaking it down into smaller parts is not efficient enough, as we need more detail to get added value from the data and determine specific action items.

For these reasons, it is important for utility professionals to have a system where they can analyze the grid alerts and warnings geospatially. This capability unlocks the possibilities for further producing insights from the data.

The Data Solutions and Methods

Awesense Energy Data Model (EDM) is ready for such analysis.

In general, two types of data are required:

1. The geospatial and connectivity data of the grid (GIS network model).
2. Time series data from measurement sensors in the grid for analysis.

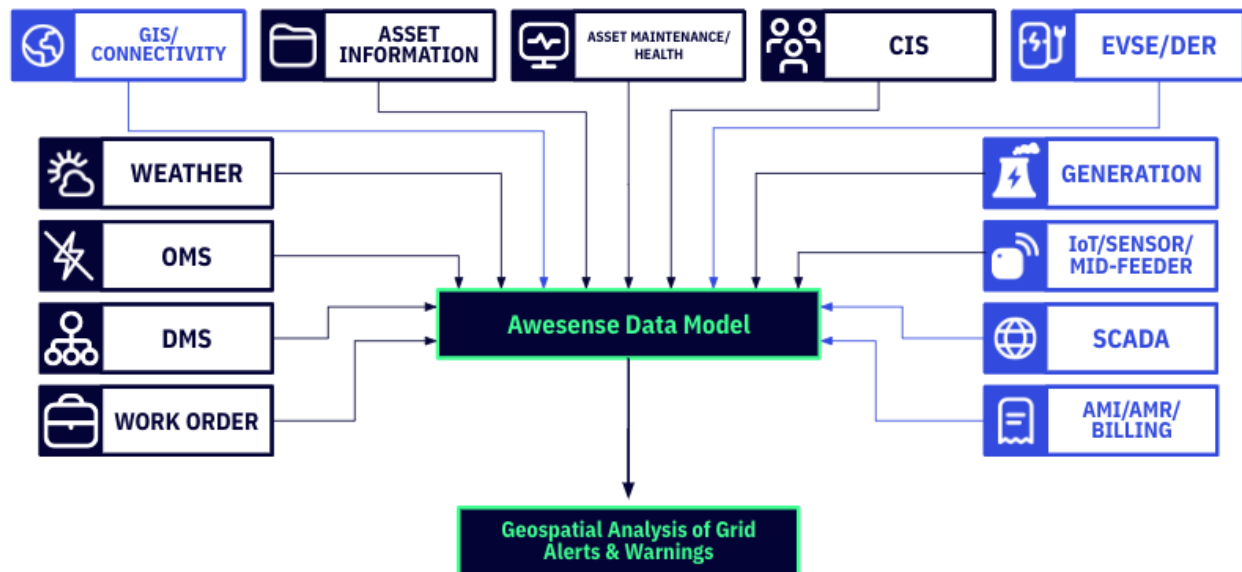
Once the time series are ingested and/or integrated with Awesense Platform, threshold values can be set for any measured quantities - e.g. current, voltage, power factor, etc., but also for parameters about the time-series-generating device (sensor) - e.g. communication problem, battery status, etc. This is secured through True Grid Intelligence's (TGI) Grid Alerts feature.

After the alerts and warnings are generated and stored within the Energy data model, the Awesense Platform offers a very easy way of executing the geospatial analysis of grid alerts and warnings. The unique integration of geospatial data (GIS) with time series data in the Awesense Platform enables this type of analysis. Such geospatial analysis can be conducted using the TGI Grid Alerts page, or analysts can build their own analysis using EDM's API interfaces.

For building specialized analysis using the EDM APIs, the pre-built tracking functions speed up the creation of analyses for selected parts of the grid. The *grid_get_downstream()* function returns a list of *grid_elements* through which the current flows from the selected element. This makes it very easy to detect all the measurement sensors, for example, on a feeder or for a station or area. Alerts and warnings are assigned to sensors, and by combining these two data sources, we can therefore determine the number of Alerts and Warnings for each feeder, station or area in the grid. This data is then displayed on a map and in a table for easy interpretation. This is enabled by the trace and export functions based on the tracing that is part of the Awesense Platform. Without topology data, we would not be able to analyze smaller parts of the grid and get into the detail needed for such analyses.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.